[SUBMIT ANSWERS TO ALL QUESTIONS]

Curves in parametric form. Tangent, normal and binormal vectors.

The length element of a curve $\boldsymbol{r}(t) = x(t)\boldsymbol{i} + y(t)\boldsymbol{j} + z(t)\boldsymbol{k}$ is $ds = |\boldsymbol{r}'(t)|dt$, where $\boldsymbol{r}'(t) \equiv d\boldsymbol{r}/dt$. The arc length between $t = a$ and $t = b$ is

$$s = \int_a^b |\boldsymbol{r}'(t)|dt = \int_a^b \sqrt{(dx/dt)^2 + (dy/dt)^2 + (dz/dt)^2}\, dt.$$

Three unit vectors, $\boldsymbol{T}$ (tangent), $\boldsymbol{N}$ (normal) and $\boldsymbol{B}$ (binormal) for a curve in 3D are defined by

$$\boldsymbol{T} = \frac{d\boldsymbol{r}}{ds}, \qquad \frac{d\boldsymbol{T}}{ds} = \kappa \boldsymbol{N}, \qquad \boldsymbol{B} = \boldsymbol{T} \times \boldsymbol{N},$$

where κ is the *curvature* and $R = 1/\kappa$ is the *radius of curvature*. For a plane curve, $\boldsymbol{B} = \text{const}$, while in general, $\dfrac{d\boldsymbol{B}}{ds} = -\tau \boldsymbol{N}$, where τ is the *torsion*, and $\dfrac{d\boldsymbol{a}}{ds} \equiv \dfrac{1}{|\boldsymbol{r}'(t)|}\dfrac{d\boldsymbol{a}}{dt}$ for any vector $\boldsymbol{a}$.

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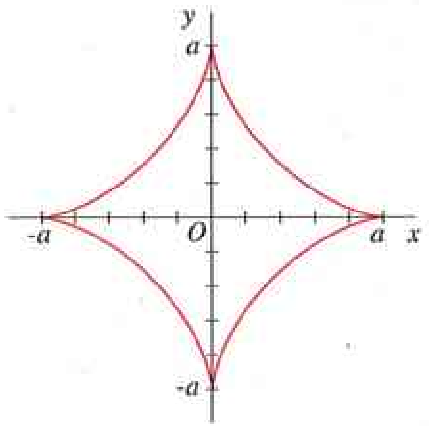
1. (a) Use integration by parts to show that [10]

$$\int \sqrt{1+t^2}\, dt = \frac{1}{2}t\sqrt{1+t^2} + \frac{1}{2}\ln\left(t + \sqrt{1+t^2}\right) + C.$$

[Hint: after integration by parts, aim to get $-\int \sqrt{1+t^2}\, dt$ on the right-hand side.]

(b) Find the arc length of the parabola $x = t$, $y = \frac{1}{2}t^2$ between points $t = 0$ and $t = 1$. [10]

2.



The curve shown is the *astroid*,

$$x^{2/3} + y^{2/3} = a^{2/3}.$$

Verify that it can be parameterised as [20]

$$x = a\cos^3 t, \quad y = a\sin^3 t, \quad 0 \le t \le 2\pi.$$

and find its arc length.

[Hint: the astroid consists of four identical segments.]

3. Find the arc lengths of the following space curves:

(a) $x = t$, $y = t^2$, $z = \frac{2}{3}t^3$, between points $t = 0$ and $t = 2$. [15]

(b) $x = e^t\cos t$, $y = e^t \sin t$, $z = e^t$, between points $t = 0$ and $t = t_0$. [15]

4. Find the tangent, normal and binormal vectors for the curve

$$x = e^t\cos t, \quad y = e^t\sin t, \quad z = e^t,$$

and determine its curvature κ, radius of curvature R and torsion τ. [30]

Problem Sheet 17 MTH1021 Solutions

Q1.

(a) Using integration by parts

$$\left[\int u\,dv = uv - \int v\,du \right].$$

$\int \sqrt{1+t^2}\,dt$ let $u = \sqrt{1+t^2}$

$$du = \frac{1}{2}(1+t^2)^{-1/2} . 2t$$

$$= \frac{t}{\sqrt{1+t^2}}.$$

$$dv = 1$$

$$v = t.$$

$$\therefore \int \sqrt{1+t^2}\,dt = t\sqrt{1+t^2} - \int \frac{t^2}{\sqrt{1+t^2}}\,dt$$

$$= t\sqrt{1+t^2} - \int \frac{t^2+1-1}{\sqrt{1+t^2}}\,dt$$

$$= t\sqrt{1+t^2} - \int \frac{t^2+1}{\sqrt{1+t^2}}\,dt + \int \frac{1}{\sqrt{1+t^2}}\,dt$$

$$= t\sqrt{1+t^2} - \underbrace{\int \sqrt{1+t^2}\,dt}_{\text{same as the original integral.}} + \int \frac{dt}{\sqrt{1+t^2}}.$$

Moving the second term to the left hand side and using the following standard integral for the last term

$$\int \frac{dt}{\sqrt{1+t^2}} = \ln\left(t + \sqrt{1+t^2}\right) + C$$

long logarithm.

we obtain

$$2\int \sqrt{1+t^2}\,dt = t\sqrt{1+t^2} + \ln\left(t+\sqrt{1+t^2}\right) + C$$

$$\therefore \int \sqrt{1+t^2}\,dt = \frac{1}{2}t\sqrt{1+t^2} + \frac{1}{2}\ln\left(t+\sqrt{1+t^2}\right) + C.$$

where C is an arbitrary constant.

$\boxed{10}$

(b) $x = t$, $y = \frac{1}{2}t^2$

$x'(t) = 1$, $y'(t) = t$.

and the element of length is

$$ds = \sqrt{(x'(t))^2 + (y'(t))^2}\,dt$$

with no z- component as the curve lies in the x-y plane.

$$ds = \sqrt{1+t^2}\,dt$$

Hence the arc length of the curve between the points $t=0$ and $t=1$ is given by

$$S = \int_0^1 ds = \int_0^1 \sqrt{1+t^2}\,dt.$$

$$= \frac{1}{2}\left[t\sqrt{1+t^2} + \ln\left(t+\sqrt{1+t^2}\right)\right]_0^1$$

using the integral from part (a).

$$S = \frac{1}{2}\left[1\sqrt{2} + \ln\left(1+\sqrt{2}\right) - 0 - \underbrace{\ln 1}_{0}\right]$$

$$\Rightarrow S = \frac{1}{2}\left[\sqrt{2} + \ln\left(1+\sqrt{2}\right)\right].$$

$\boxed{10}$.

Q2.

$$x = a\cos^3 t \quad , y = a\sin^3 t \quad \ldots (*)$$

and $$x^{2/3} + y^{2/3} = a^{2/3} \quad \ldots (**)$$

Substituting $(*)$ into the left-hand-side of $(**)$ we find

$$(a\cos^3 t)^{2/3} + (a\sin^3 t)^{2/3}$$
$$= a^{2/3}(\cos^3 t)^{2/3} + a^{2/3}(\sin^3 t)^{2/3}$$
$$= a^{2/3}\cos^2 t + a^{2/3}\sin^2 t$$
$$= a^{2/3}\underbrace{[\cos^2 t + \sin^2 t]}_{=1}$$
$$= a^{2/3} \quad \text{as required.}$$

As the parameter t increases from 0 to 2π, the point (x,y) slides along the curve

For

$t=0$ $\quad x=a \;, y=0$

$t=\pi/2$ $\quad x=0 \;, y=a$

$t=\pi$ $\quad x=-a, \; y=0$

$t=\frac{3\pi}{2}$ $\quad x=0 \quad y=-a$

(the first four: Four cusps of the astroid.)

$t=2\pi$ $\quad x=a \quad y=0$ (back to the beginning)

The astroid consists of 4 identical segments, so we can find its total length by finding the arc length of one segment, say in the first quadrant, between $t=0$ and $t=\pi/2$ and multiplying it by 4.

$x = a\cos^3 t$, $y = a\sin^3 t$

$x'(t) = -3a\cos^2 t \sin t$ $y'(t) = 3a\sin^2 t\cos t$.

Hence the element of length is

$$ds = \sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt.$$

$$= \sqrt{9a^2\cos^4 t\sin^2 t + 9a^2\sin^4 t\cos^2 t}\, dt$$

$$= 3a\sqrt{\cos^2 t\sin^2 t\,\underbrace{(\cos^2 t + \sin^2 t)}_{=1}}\, dt$$

$$= 3a\sqrt{\cos^2 t\sin^2 t}\, dt$$

and the length of the astroid is

$$s = 4 \times 3a\int_0^{\pi/2}\sqrt{\cos^2 t\sin^2 t}\, dt.$$

$$= 12a\int_0^{\pi/2}\cos t\sin t\, dt$$

Note $\sqrt{\cos^2 t\sin^2 t} = |\cos t\sin t| = \cos t\sin t$ for $0 \le t \le \pi/2$

since $\cos t \geq 0$ and $\sin t \geq 0$ for this range of t.

$$\therefore s = 12a\int_0^{\pi/2}\sin t\, d(\sin t)$$

$$= 12a\left[\frac{\sin^2 t}{2}\right]_0^{\pi/2}.$$

$$= 6a[1-0] = 6a$$

Note that each of the four segments of the astroid is $1.5a$, slightly longer than the straight line segments connecting its 'vertices', eg $(0,a)$ and $(a,0)$, which equal $\sqrt{2}\,a \approx 1.41a$. [20]

Q3.

(a) $x = t \quad , y = t^2 \quad , \quad z = 2/3\, t^3$

$x'(t) = 1 \quad , \quad y'(t) = 2t \quad , \quad z'(t) = 2t^2$

and the length element is

$$ds = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\, dt$$

$$= \sqrt{1 + 4t^2 + 4t^4}\quad dt.$$

and the arc length of the curve between the points $t=0$ and $t=2$ is

$$s = \int_0^2 ds = \int_0^2 \sqrt{1+4t^2+4t^4}\, dt$$

$$= \int_0^2 \sqrt{(1+2t^2)^2}\quad dt$$

$$= \int_0^2 (1+2t^2)\, dt$$

$$= \left[t + \frac{2}{3}t^3\right]_0^2$$

$$= 2 + \frac{2}{3}\cdot 8 - 0$$

$$= 2 + \frac{16}{3} = 2 + 5\tfrac{1}{3} = 7\tfrac{1}{3}. \quad [15].$$

(b) $x = e^t \cos t \quad , \quad x'(t) = e^t(\cos t - \sin t)$

$y = e^t \sin t \quad , \quad y'(t) = e^t(\sin t + \cos t)$

$z = e^t \quad , \quad z'(t) = e^t$

and the length element is

$$ds = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\quad dt$$

$$ds = \sqrt{e^{2t}(\cos t - \sin t)^2 + e^{2t}(\sin t + \cos t)^2 + e^{2t}}\, dt$$

$$= \sqrt{e^{2t}\left[\cos^2 t + \sin^2 t - \cancel{2\cos t \sin t} + \sin^2 t + \cos^2 t + \cancel{2\cos t \sin t} + 1\right]}\, dt$$

$$= e^t \sqrt{1 + 1 + 1}\, dt$$

$$= e^t \sqrt{3}\, dt$$

and the arc length of the curve between points $t=0$ and $t=t_0$ is

$$s = \int_0^{t_0} ds = \int_0^{t_0} e^t \sqrt{3}\, dt$$

$$= \sqrt{3}\left[e^t\right]_0^{t_0}$$

$$= \sqrt{3}\left[e^{t_0} - 1\right).$$

[15].

Q4.

$x = e^t \cos t$, $y = e^t \sin t$, $z = e^t$.

can be written in vector form as

$$\underline{r}(t) = (e^t \cos t, e^t \sin t, e^t)$$

Then

$$\underline{r}'(t) = \left[e^t(\cos t - \sin t), e^t(\sin t + \cos t), e^t\right]$$

$$= e^t\,(\cos t - \sin t, \sin t + \cos t, 1)$$

↑ common factor.

$$|\underline{r}'(t)| = e^t \sqrt{(\cos t - \sin t)^2 + (\sin t + \cos t)^2 + 1}$$

$$= \sqrt{3}\, e^t \quad \text{after cancellations}$$

Hence, the <u>tangent</u> vector is

$$\underline{T} = \frac{d\underline{r}}{ds} = \frac{1}{|\underline{r}'(t)|}\frac{d\underline{r}}{dt} = \frac{\underline{r}'(t)}{|\underline{r}'(t)|}$$

$$\Rightarrow \underline{T} = \frac{e^t(\cos t - \sin t, \sin t + \cos t, 1)}{\sqrt{3}\, e^t}$$

$$\Rightarrow \underline{T} = \frac{1}{\sqrt{3}}\left(\cos t - \sin t, \sin t + \cos t, 1\right)$$

re-writing

$$\frac{d\underline{T}}{ds} = \kappa \underline{N} \quad \text{as} \quad \frac{1}{|\underline{r}'(t)|}\frac{d\underline{T}}{dt} = \kappa \underline{N}$$

we obtain for the left-hand-side.

$$\frac{1}{\sqrt{3}e^t}\frac{1}{\sqrt{3}}\left(-\sin t - \cos t, \cos t - \sin t, 0\right) \quad \ldots (*)$$

Since $|\underline{N}| = 1$, $\left|\frac{d\underline{T}}{ds}\right| = \kappa$

Taking the modulus (length) of the above vector, we find

$$\kappa = \frac{1}{3}e^{-t}\sqrt{(-\sin t - \cos t)^2 + (\cos t - \sin t)^2}$$

$$= \frac{e^{-t}}{3}\sqrt{\sin^2 t + \cos^2 t + 2\cancel{\sin t \cos t} + \cos^2 t + \sin^2 t - 2\cancel{\cos t \sin t}}$$

$$= \frac{e^{-t}}{3}\sqrt{1 + 1}$$

$$\kappa = \frac{\sqrt{2}}{3}e^{-t} \qquad \text{(curvature)} \quad \ldots (**)$$

and the radius of curvature is

$$R = \frac{1}{K} = \frac{3}{\sqrt{2}} e^{t}.$$

Equation (*) above is equal to $K\underline{N}$, and using (**) we have.

$$\frac{1}{3} e^{-t}(-\sin t - \cos t, \cos t - \sin t, 0)$$

$$= \underbrace{\frac{\sqrt{2}}{3} e^{-t}}_{K} \cdot \underbrace{\frac{1}{\sqrt{2}}(-\sin t - \cos t, \cos t - \sin t, 0)}_{\underline{N}}$$

hence the normal vector $\underline{N}$ is given by

$$\underline{N} = \frac{1}{\sqrt{2}}(-\sin t - \cos t, \cos t - \sin t, 0).$$

To find the binormal we use

$$\underline{B} = \underline{T} \times \underline{N} = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \cos t - \sin t & \sin t + \cos t & 1 \\ -\sin t - \cos t & \cos t - \sin t & 0 \end{vmatrix}$$

$$= \frac{1}{\sqrt{6}} \left[\underline{i} \begin{vmatrix} \sin t + \cos t & 1 \\ \cos t - \sin t & 0 \end{vmatrix} - \underline{j} \begin{vmatrix} \cos t - \sin t & 1 \\ -\sin t - \cos t & 0 \end{vmatrix} \right.$$

$$\left. + \underline{k} \begin{vmatrix} \cos t - \sin t & \sin t + \cos t \\ -\sin t - \cos t & \cos t - \sin t \end{vmatrix} \right]$$

$$= \frac{1}{\sqrt{6}} \left[-\underline{i}(\cos t - \sin t) + \underline{j}(-\sin t - \cos t) \right.$$
$$\left. + \underline{k}\left((\cos t - \sin t)^2 + (\sin t + \cos t)^2\right) \right]$$

$$= \frac{1}{\sqrt{6}} \left[\underline{i}(\sin t - \cos t) + \underline{j}(-\sin t - \cos t) + 2\underline{k} \right]$$

$$\Rightarrow \quad \underline{B} = \frac{1}{\sqrt{6}}\left(\sin t - \cos t\,,\, -\sin t - \cos t\,,\, 2\right)$$

is the binormal.

Finally for the torsion

$$\frac{d\underline{B}}{ds} = -\tau\,\underline{N} \qquad \text{or}$$

$$\frac{1}{|\underline{r}'(t)|}\,\frac{d\underline{B}}{dt} = -\tau\,\underline{N}$$ we obtain for the L.H.S.

$$\frac{1}{\sqrt{3}e^{t}}\,\frac{1}{\sqrt{6}}\left(\cos t + \sin t,\, -\cos t + \sin t\,,\, 0\right)$$

$$\frac{1}{3e^{t}}\,(-1)\,\underbrace{\frac{1}{\sqrt{2}}\left(-\sin t - \cos t\,,\, \cos t - \sin t\,,\, 0\right)}_{\underline{N} \text{ from before}}$$

Comparing this with the R.HS. $-\tau\,\underline{N}$ we see that the torsion τ is given by

$$\boxed{\tau = \frac{e^{-t}}{3}.}$$

[30].